GAME THEORY & NASH EQUILIBRIA — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 16, 2026

Diagram-based questions on Game Theory & Nash Equilibria

Question 1

10.0 points

Analyze the extensive form game tree in the diagram below for a sequential two-player game where Player 1 can choose between actions A and B, and Player 2 can then choose between X and Y after observing Player 1's action. Identify the subgame perfect Nash equilibrium by performing backward induction on this game tree.

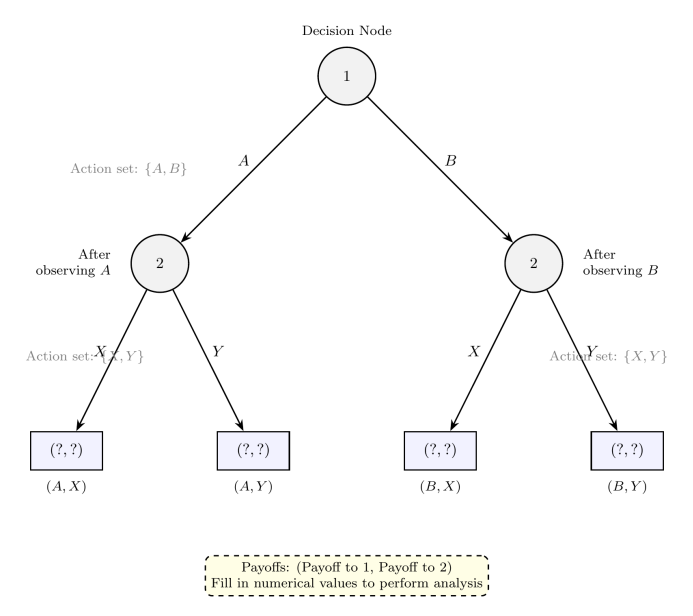

Fig. 1. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: If Player, Identify, Nash, Player, Backward, nash equilibrium; The diagram contains the label "Player 1", but the word "Player" is on the forbidden labels list, constituting an answer leak.. A game tree for a sequential two-player game IMPORTANT: Values or sequences in the diagram don't match the question. Specifically: The payoffs for all four outcomes are missing and are represented as "(?, ?)", making it impossible for the. The question requires the student to perform backward induction, which is impossible without knowing the numerical payoffs at the terminal nodes, but the diagram only shows placeholders "(?,?)".. Regenerate using EXACTLY the values and sequences specified.

Question 2

10.0 points

Consider the payoff matrix for a Prisoner's Dilemma shown in the diagram below: | | Cooperate | Defect | |-------|----------|--------| | C | (3,3) | (0,5) | | D | (5,0) | (1,1) | Identify all Nash equilibria in this game and explain why they are Nash equilibria.

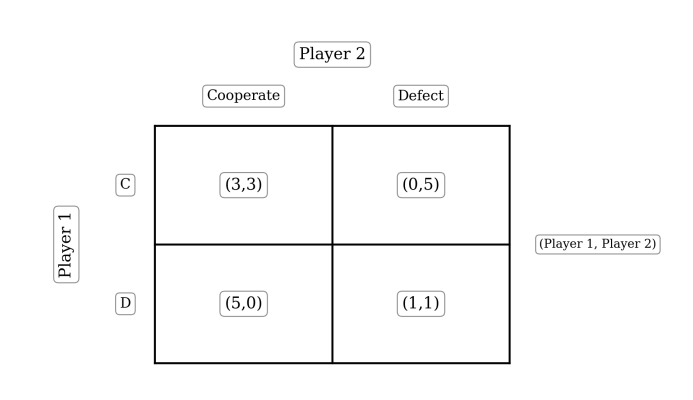

Fig. 2. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Nash, Defect, all Nash equilibria in this game, why they are Nash equilibria, nash equilibrium, dominant strategy; The diagram includes the label "Defect", which is on the forbidden labels list and is a key part of the answer the student must identify.; The diagram includes the label "Defect", which is on the question-specific forbidden labels list.. Remove the label 'Defect'. The diagram should show a payoff matrix for a two-player

Question 3 10.0 points

Evaluate a mixed strategy Nash equilibrium in a coordination game given by the payoff table in the diagram below, where both players choose high or low effort: | | High | Low | |----------|-------|-------| | High | (2,2) | (0,3) | | Low | (3,0) | (1,1) | Use the mixed strategy method to find the equilibria, and determine the probabilities with which each player should randomize their strategies.

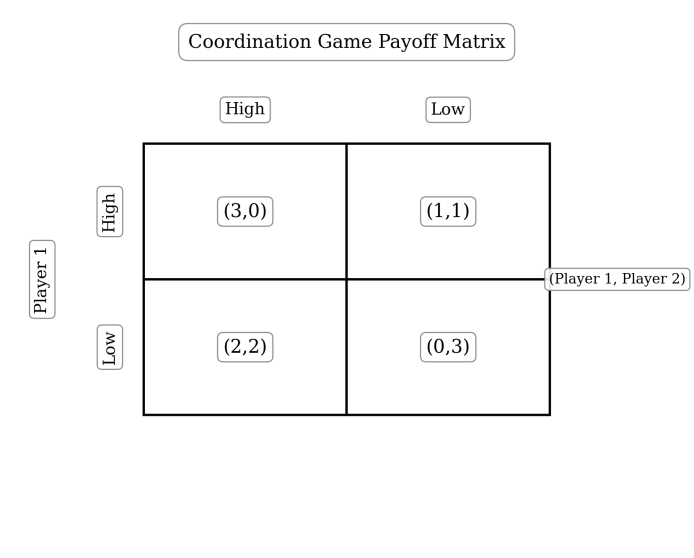

Fig. 3. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Let Player, For Player, Let, Player, High, Referring, Set, Mixed, Nash, equilibria; The diagram contains the labels "Player" and "High", which are explicitly listed in both the general and question-specific forbidden labels lists.; The diagram contains the labels "Player" and "High", which are explicitly listed in the forbidden labels list for this question.. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Let Player, For Player, Let, Player, High, Referring, Set, Mixed, Nash, equilibria; The diagram contains the labels "Player" and "High", which are explicitly listed in both the general and question-specific forbidden labels lists.. A payoff matrix for a coordination game where both players choose high or low effort. The matrix has two rows and two columns. The first row and first column are labeled 'High', and the second row and column are labeled 'Low'. The payoffs are as follows: (2,2) for High-High, (0,3) for High-Low, (3,0) for Low-High, and (1,1) for Low-Low. Each cell shows the payoff for Player 1 and Player 2. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Let Player • For Player • Let • Player • High • Referring • Set • Mixed • Nash • equilibria • syndrome table • nash equilibrium • dominant strategy • equal areas The diagram will FAIL review if any of these appear. IMPORTANT correction needed: The diagram contains the labels "Player" and "High", which are explicitly listed in both the general and question-specific forbidden labels lists.. Specifically: The labels "Player 1", "(Player 1, Player 2)", and "High" are present, but "Player" and ". Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT correction needed: The diagram contains the labels "Player" and "High", which are explicitly listed in the forbidden labels list for this question.. Specifically: The diagram includes the labels "Player 1", "(Player 1, Player 2)", and "High", which contain the forbidden terms "Player" and. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 4

10.0 points

Use backward induction on the game tree depicted in the diagram below for a three-stage sequential game with perfect information. Each player has one possible optimal move at each decision node. Identify the optimal strategy for each player, and explain how backward induction assists in determining this strategy.

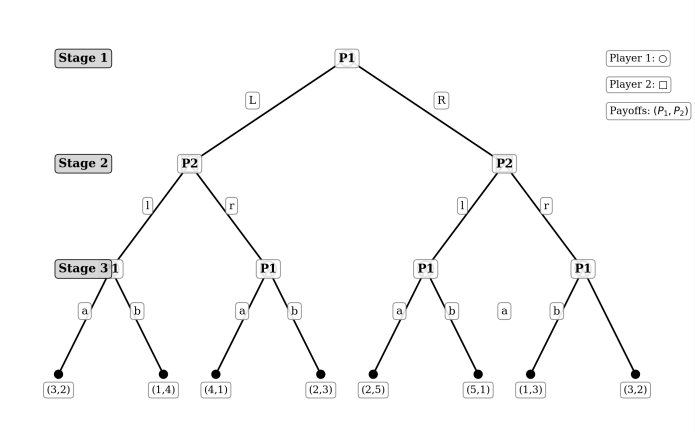

Fig. 4. A three-stage sequential game tree with perfect information, showing decision nodes for Player 1 and Player 2, and end nodes representing outcomes. Each player has one optimal move at each decision node. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Determine • Use • Finally • Backward • optimal strategy for each player • optimal • dominant strategy The diagram will FAIL review if any of these appear.

Question 5 10.0 points

Given the Pareto frontier in the diagram below illustrating efficient vs inefficient outcomes, identify the Pareto efficient points and describe the trade-offs at inefficient points. Explain how any shift in the curve can alter efficiency in allocation.

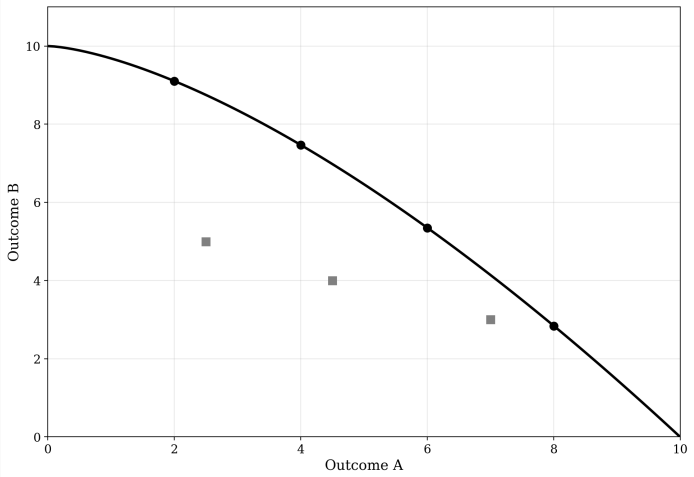

Fig. 5. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Pareto, Pareto efficient points, trade-offs at inefficient points, pareto frontier, pareto front, feasible region; The diagram explicitly labels points as "Efficient" and "Inefficient", which directly answers the question asking the student to identify these categories of points.; The diagram explicitly labels points as "Efficient" and "Inefficient", which directly answers the first part of the question asking the student to identify these points.. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Pareto, Pareto efficient points, trade-offs at inefficient points, pareto frontier, pareto front, feasible region; The diagram explicitly labels points as "Efficient" and "Inefficient", which directly answers the question asking the student to identify these categories of points.. A 2D plot with the x-axis labeled 'Outcome A' and the y-axis labeled 'Outcome B'. A smooth, downward-sloping curve starts high on the y-axis and ends far out on the x-axis. There are several points, marked with circles, located directly on this curve. There are also several points, marked with squares, located in the area below the curve. An arrow should point from the original curve to a new, dashed curve located further out from the origin, indicating an outward shift. Do not label any points or regions as 'Efficient' or 'Inefficient'. IMPORTANT correction needed: The diagram explicitly labels points as "Efficient" and "Inefficient", which directly answers the first part of the question asking the student to identify these points.. Specifically: The labels "Efficient" and "Inefficient" are present, which constitutes an answer leak. The diagram is also missing the required. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.